

CONSTRUCTION OS

ASSETS MODULE — FULL PRODUCT & SYSTEM ARCHITECTURE

1. Module Purpose

The Assets Module is the central system of record for every physical, operational, technical, leased, owned, assigned, mobile, fixed, consumable-linked, and digitally identifiable asset used by the company.

The module must answer:

- What assets does the company control?
- Who owns each asset?
- Where is each asset located?
- Which company, project, building, site, department, employee, warehouse, or client is responsible for it?
- Is the asset available, assigned, operating, under repair, missing, retired, or disposed?
- What is its current condition?
- What has it cost?
- What maintenance is due?
- What documents belong to it?
- What work has been performed on it?
- What is its depreciation, book value, replacement value, and lifecycle history?
- Which asset was used on which project, task, site activity, or cost centre?
- Which risks, inspections, incidents, warranties, permits, or compliance requirements are linked to it?

The module must operate as an enterprise asset management layer across the entire Construction OS.

2. Core Architectural Principle

Every asset must exist as a unique platform record with:

- A permanent Asset ID
- A human-readable asset code
- A complete lifecycle
- A responsible company
- A responsible owner or custodian
- A current location
- A current operational status
- A complete transaction history
- Linked documents
- Linked financial records

- Linked maintenance records
- Linked inspections
- Linked assignments
- Linked projects and cost centres
- A complete audit trail

An asset may change location, project, employee, condition, value, ownership status, or operational status, but its historical record must never be overwritten.

3. Asset Scope

The Assets Module must support the following asset families.

3.1 Heavy Machinery

Examples:

- Excavators
- Bulldozers
- Cranes
- Loaders
- Graders
- Rollers
- Concrete pumps
- Piling rigs
- Asphalt equipment
- Drilling machines
- Tower cranes
- Mobile cranes

3.2 Vehicles and Fleet

Examples:

- Cars
- Vans
- Trucks
- Dump trucks
- Trailers
- Buses
- Motorcycles
- Special-purpose vehicles
- Service vehicles

3.3 Site Equipment

Examples:

- Generators
- Compressors
- Mixers
- Vibrators
- Pumps
- Scaffolding systems
- Hoists
- Welding equipment
- Lighting towers
- Temporary site installations

3.4 Tools

Examples:

- Power tools
- Hand tools
- Measuring tools
- Surveying devices
- Laser equipment
- Testing instruments

3.5 IT and Technology Assets

Examples:

- Computers
- Laptops
- Servers
- Monitors
- Mobile phones
- Tablets
- Printers
- Routers
- Networking equipment
- Cameras
- Drones
- Software-linked devices
- Access-control equipment

3.6 Office Assets

Examples:

- Furniture
- Appliances

- Meeting-room equipment
- Air-conditioning equipment
- Security systems
- Office installations

3.7 Property and Fixed Assets

Examples:

- Land
- Offices
- Warehouses
- Apartments
- Commercial units
- Site compounds
- Workshops
- Storage yards
- Buildings
- Infrastructure installations

3.8 Safety and Compliance Assets

Examples:

- Fire extinguishers
- First-aid equipment
- Harnesses
- Gas detectors
- Safety barriers
- Emergency equipment
- Personal protective equipment requiring individual tracking

3.9 Rented or Leased Assets

Assets controlled but not owned by the company.

Examples:

- Rented machinery
- Leased vehicles
- Temporary equipment
- Client-owned assets
- Supplier-owned assets
- Subcontractor-provided assets

3.10 Client or Third-Party Assets

Assets managed, operated, maintained, stored, inspected, or tracked by the company but owned by another legal entity.

3.11 Asset Groups and Component Assets

An asset may contain sub-assets or components.

Examples:

- Vehicle with GPS device and specialised equipment
 - Crane with motor, boom, hook block, and control unit
 - Server rack containing individual servers
 - Building containing HVAC, elevators, generators, and fire systems
 - Scaffolding system containing multiple trackable sections
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4. Asset Classification Architecture

Every asset must be classified using a configurable hierarchy.

4.1 Classification Levels

- Asset Domain
- Asset Category
- Asset Subcategory
- Asset Type
- Asset Model
- Asset Variant
- Asset Component Type

Example:

Fleet

→ Heavy Vehicles

→ Dump Trucks

→ 6x4 Dump Truck

→ Mercedes Arocs

→ 4145K Variant

4.2 Classification Templates

Each asset type may define:

- Required fields
- Optional fields
- Measurement units
- Inspection templates
- Maintenance schedules
- Depreciation rules
- Compliance rules
- Document requirements
- Assignment restrictions

- Operating requirements
- Custom fields
- Asset status rules

4.3 Custom Fields

Administrators may create custom asset fields by:

- Company
- Asset category
- Asset type
- Project type
- Country
- Regulatory requirement

Supported field types:

- Text
- Number
- Currency
- Date
- Date and time
- Yes/No
- Dropdown
- Multi-select
- User
- Company
- Project
- Location
- Asset reference
- Document reference
- Formula
- Measurement
- Serial identifier
- URL

5. Asset Identity

Each asset record must contain a unique identity layer.

5.1 System Identity

- Internal UUID
- Asset code
- Asset name
- Asset category
- Asset type
- Parent asset
- Component assets

- Company
- Corporate group
- Legal owner
- Operational owner
- Responsible department
- Cost centre
- Accounting code

5.2 Manufacturer Identity

- Manufacturer
- Brand
- Model
- Variant
- Serial number
- VIN
- Chassis number
- Engine number
- Registration number
- Production year
- Purchase year
- Country of origin

5.3 Tracking Identity

- Barcode
- QR code
- NFC identifier
- RFID identifier
- GPS device identifier
- IoT device identifier
- External system code
- Supplier code
- Legacy asset code

The same identifier cannot be assigned to multiple active assets unless specifically permitted for grouped assets.

6. Asset Ownership Model

The platform must distinguish between:

- Legal owner
- Paying company
- Operating company
- Custodian
- Responsible department
- Responsible employee
- Current user

- Client owner
- Supplier owner
- Leasing company
- Insurance holder
- Maintenance provider

Supported ownership types:

- Company-owned
- Group-owned
- Jointly owned
- Leased
- Rented
- Financed
- Client-owned
- Supplier-owned
- Subcontractor-owned
- Employee-owned
- Government-owned
- Temporarily controlled
- Managed on behalf of another entity

Ownership changes must generate historical records.

7. Asset Lifecycle

Every asset must follow a controlled lifecycle.

7.1 Standard Lifecycle

1. Requested
2. Planned
3. Approved
4. Ordered
5. In procurement
6. Awaiting delivery
7. Received
8. Under inspection
9. Registered
10. Available
11. Assigned
12. In operation
13. Temporarily inactive
14. Under maintenance
15. Under repair
16. Quarantined
17. Damaged
18. Lost
19. Stolen

- 20. Returned
- 21. Stored
- 22. Transferred
- 23. For sale
- 24. Retired
- 25. Disposed
- 26. Sold
- 27. Scrapped
- 28. Archived

7.2 Status Architecture

The module must separate:

- Lifecycle status
- Operational status
- Availability status
- Condition status
- Compliance status
- Maintenance status
- Financial status

Example:

- Lifecycle: Active
 - Operational: Under repair
 - Availability: Unavailable
 - Condition: Poor
 - Compliance: Inspection expired
 - Maintenance: Overdue
 - Financial: Fully depreciated
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8. Asset Record Architecture

Each asset record must include the following sections.

8.1 Overview

- Asset image
- Asset name
- Asset code
- Asset category
- Status
- Condition
- Availability
- Current company
- Current project
- Current location
- Responsible person

- Ownership type
- Critical warnings
- Upcoming actions
- Current operating metrics

8.2 Technical Information

- Manufacturer
- Model
- Serial number
- Specifications
- Capacity
- Dimensions
- Weight
- Power
- Fuel type
- Energy type
- Operating limits
- Technical parameters
- Custom technical fields

8.3 Ownership and Financial Information

- Purchase price
- Purchase date
- Supplier
- Invoice
- Contract
- Financing details
- Lease details
- Initial value
- Current book value
- Depreciation method
- Depreciation period
- Salvage value
- Replacement value
- Insurance value
- Disposal value
- Total lifecycle cost

8.4 Assignment

- Assigned company
- Assigned department
- Assigned project
- Assigned site
- Assigned employee
- Assigned team
- Assigned cost centre
- Start date
- Expected return date

- Actual return date
- Assignment status

8.5 Location

- Current location
- Previous locations
- GPS coordinates
- Geofenced site
- Building
- Floor
- Room
- Warehouse
- Storage zone
- Yard position
- Map view

8.6 Maintenance

- Maintenance plan
- Service intervals
- Last service
- Next service
- Meter at service
- Maintenance provider
- Maintenance history
- Active work orders
- Open defects
- Spare parts used
- Downtime
- Cost history

8.7 Inspections and Compliance

- Inspection plan
- Inspection history
- Certificates
- Licences
- Permits
- Calibration
- Insurance
- Registration
- Expiration dates
- Compliance status
- Failed checks
- Corrective actions

8.8 Documents

- Purchase documents
- Invoices

- Contracts
- Manuals
- Warranties
- Certificates
- Registration documents
- Insurance
- Inspection reports
- Photos
- Drawings
- Maintenance reports
- Handover forms
- Disposal records

8.9 Activity and Audit

- Status changes
 - Assignments
 - Transfers
 - Inspections
 - Maintenance
 - Document uploads
 - Financial changes
 - Location changes
 - User actions
 - System automations
 - API actions
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9. Asset Location Architecture

The module must support a universal location hierarchy.

9.1 Location Types

- Country
- Region
- City
- Company office
- Project
- Construction site
- Building
- Block
- Structure
- Floor
- Zone
- Room
- Warehouse
- Warehouse zone
- Storage rack

- Yard
- Vehicle
- Temporary site
- External provider
- Employee possession
- Client location
- Unknown location

9.2 Location History

Every movement must record:

- Origin
- Destination
- Date and time
- Initiated by
- Approved by
- Transport method
- Transport reference
- Reason
- Condition before transfer
- Condition after transfer
- Handover document
- Receiving person
- Photos
- Notes

10. Asset Assignment Architecture

Assets may be assigned to:

- Employee
- Team
- Department
- Company
- Project
- Site
- Building
- Unit
- Task
- Work package
- Cost centre
- Vehicle
- Warehouse
- Client
- Subcontractor

10.1 Assignment Rules

The system must support:

- Temporary assignments
- Permanent assignments
- Shared assignments
- Exclusive assignments
- Scheduled future assignments
- Recurring assignments
- Project-based assignments
- Shift-based assignments

10.2 Handover Process

1. Assignment request
2. Availability check
3. Approval
4. Condition inspection
5. Handover
6. Acceptance
7. Active usage
8. Return request
9. Return inspection
10. Closure

The handover may require:

- Signature
 - Photos
 - Meter reading
 - Fuel level
 - Condition checklist
 - Accessories checklist
 - Documents
 - Responsible-party confirmation
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11. Reservations and Availability

Users must be able to reserve assets.

11.1 Reservation Record

- Asset or asset type
- Requesting user
- Project
- Task
- Location

- Start date
- End date
- Quantity
- Purpose
- Priority
- Approval
- Delivery requirements

11.2 Reservation Conflict Rules

The system must detect:

- Date conflicts
- Maintenance conflicts
- Location conflicts
- Compliance restrictions
- Capacity restrictions
- Employee certification restrictions
- Project restrictions
- Asset condition restrictions

11.3 Availability States

- Available
- Reserved
- Partially reserved
- Assigned
- In transit
- Under maintenance
- Unavailable
- Out of service
- Unknown

12. Maintenance Management

The Assets Module must include a complete maintenance architecture.

12.1 Maintenance Types

- Preventive maintenance
- Corrective maintenance
- Predictive maintenance
- Condition-based maintenance
- Emergency maintenance
- Warranty maintenance
- Manufacturer service
- Regulatory service
- Calibration

- Cleaning
- Software update
- Safety inspection

12.2 Maintenance Triggers

Maintenance may be triggered by:

- Calendar interval
- Operating hours
- Mileage
- Cycles
- Production quantity
- Fuel consumption
- Sensor reading
- Inspection result
- Reported defect
- Manufacturer schedule
- Warranty condition
- Regulatory deadline

12.3 Maintenance Plan

Each plan contains:

- Plan name
- Asset type
- Applicable assets
- Trigger
- Frequency
- Tolerance
- Checklist
- Required skills
- Required parts
- Estimated duration
- Estimated cost
- Service provider
- Approval requirement
- Completion documents

12.4 Maintenance Work Order

- Work order number
- Asset
- Maintenance type
- Priority
- Problem description
- Diagnosis
- Assigned technician
- External provider

- Planned date
- Actual date
- Labour hours
- Parts used
- Cost
- Downtime
- Meter reading
- Photos
- Documents
- Completion result
- Approval
- Rework requirement

12.5 Defects

Users may report defects using:

- Asset page
- Mobile scan
- Inspection
- Site diary
- Task
- Incident
- Employee report
- Automated sensor alert

Defects must support:

- Severity
- Operational impact
- Safety impact
- Photos
- Immediate action
- Asset lockout
- Assigned responsible party
- Corrective task
- Maintenance order
- Root cause
- Closure approval

13. Meter and Usage Tracking

Assets may have one or more meters.

Examples:

- Kilometres
- Engine hours
- Operating hours

- Cycles
- Energy consumed
- Fuel consumed
- Output quantity
- Lift count
- Pump volume
- Runtime

13.1 Meter Records

- Meter type
- Unit
- Reading
- Reading date
- Source
- Submitted by
- Photo evidence
- Automatic/manual
- Validation status

13.2 Reading Validation

The system must identify:

- Negative progression
- Unrealistic increases
- Duplicate readings
- Missing readings
- Suspected meter reset
- Suspected data entry error

14. Fuel and Energy Management

For relevant assets, track:

- Fuel type
- Fuel capacity
- Fuel transactions
- Charging transactions
- Fuel card
- Supplier
- Quantity
- Unit price
- Total cost
- Odometer or meter reading
- Project
- Driver
- Location

- Receipt
- Consumption rate
- Variance
- Suspected misuse

Supported energy sources:

- Diesel
 - Petrol
 - LPG
 - Electricity
 - Hybrid
 - Hydrogen
 - Other configurable types
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15. Inspection Architecture

15.1 Inspection Types

- Daily pre-use inspection
- Weekly inspection
- Monthly inspection
- Annual inspection
- Safety inspection
- Technical inspection
- Return inspection
- Transfer inspection
- Regulatory inspection
- Damage inspection
- Acceptance inspection
- Disposal inspection

15.2 Inspection Template

- Sections
- Questions
- Pass/fail fields
- Measurements
- Photo requirements
- Signature requirements
- Critical questions
- Conditional questions
- Automatic defect creation
- Automatic asset lockout
- Approval rules

15.3 Inspection Outcomes

- Passed

- Passed with observations
 - Conditional pass
 - Failed
 - Unsafe
 - Requires repair
 - Requires reinspection
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16. Compliance Management

The module must track:

- Vehicle registration
- Technical inspection
- Insurance
- Equipment certification
- Calibration certificates
- Operator licence requirements
- Environmental permits
- Safety certification
- Warranty
- Rental compliance
- Road tax
- Operating permits

16.1 Compliance Status

- Valid
- Expiring soon
- Expired
- Missing
- Under review
- Suspended
- Not applicable

Expired critical compliance may automatically prevent assignment or operation.

17. Warranty Management

Warranty records must contain:

- Provider
- Start date
- Expiry date
- Coverage
- Exclusions
- Claim procedure
- Contract

- Maximum usage
- Covered components
- Contact person
- Claim history

The system must warn users before paid maintenance is approved when an asset may still be under warranty.

18. Insurance Management

- Insurance provider
 - Policy number
 - Coverage type
 - Covered value
 - Start date
 - Expiry date
 - Deductible
 - Premium
 - Covered locations
 - Covered users
 - Exclusions
 - Claims
 - Documents
 - Renewal status
-

19. Damage, Loss, Theft, and Incident Management

The module must support:

- Damage reports
- Lost asset reports
- Theft reports
- Accident reports
- Insurance claims
- Police reports
- Responsibility assessment
- Cost allocation
- Disciplinary link
- Recovery status
- Asset freeze
- Write-off recommendation

Incident records must link to:

- Asset

- Employee
 - Project
 - Location
 - Task
 - Site diary
 - HSE incident
 - Insurance
 - Documents
 - Financial impact
-

20. Asset Transfer Management

Transfers may occur between:

- Companies
- Projects
- Sites
- Warehouses
- Departments
- Employees
- Countries
- Legal entities
- Clients
- Suppliers

20.1 Transfer Workflow

1. Transfer request
 2. Availability validation
 3. Compliance validation
 4. Approval
 5. Dispatch inspection
 6. Dispatch
 7. In transit
 8. Receipt
 9. Receipt inspection
 10. Acceptance
 11. Ownership or assignment update
 12. Financial allocation
 13. Closure
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21. Asset Check-In and Check-Out

Mobile-first asset checkout must support:

- QR scan

- Barcode scan
 - NFC scan
 - Manual search
 - Bulk checkout
 - Bulk return
 - Signature
 - Photo
 - Condition check
 - Expected return
 - Automatic reminders
 - Overdue escalation
-

22. Procurement Integration

The Assets Module must connect directly to Procurement.

22.1 Asset Creation from Procurement

An asset record may originate from:

- Purchase request
- Approved budget
- Purchase order
- Goods receipt
- Supplier invoice
- Leasing contract
- Rental order

22.2 Receiving Workflow

When an asset is received:

- Create draft asset record
 - Capture serial number
 - Capture photos
 - Confirm quantity
 - Complete acceptance inspection
 - Link purchase order
 - Link invoice
 - Link warranty
 - Assign location
 - Assign asset code
 - Print QR label
 - Activate asset
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23. Inventory and Spare Parts Integration

Assets and inventory must remain separate concepts.

- Asset: individually traceable operational item
- Inventory item: quantity-based stock item
- Spare part: inventory item consumed or installed on an asset

The system must track:

- Spare parts issued to an asset
- Installed parts
- Removed parts
- Component serial numbers
- Part warranty
- Maintenance stock consumption
- Asset-related purchase demand
- Minimum spare-part stock

A component may become a separately tracked asset when required.

24. Project Integration

Each asset may be linked to:

- Project
- Site
- Building
- Structure
- Unit
- Phase
- Milestone
- Work package
- Task
- Site activity
- Cost code
- Budget line
- Site diary

The module must calculate:

- Asset days per project
- Operating hours per project
- Asset cost per project
- Fuel cost per project
- Maintenance cost per project
- Downtime per project
- Utilisation by project
- Internal rental charge

- External rental cost
-

25. Finance Integration

25.1 Financial Records

- Acquisition cost
- Freight
- Customs
- Installation
- Commissioning
- Capitalised cost
- Maintenance cost
- Fuel cost
- Insurance cost
- Tax
- Registration cost
- Rental cost
- Lease cost
- Repair cost
- Disposal cost
- Sale proceeds
- Depreciation
- Impairment
- Book value
- Replacement cost
- Total cost of ownership

25.2 Depreciation Methods

- Straight line
- Declining balance
- Double declining balance
- Units of production
- Usage-based
- Manual
- No depreciation

Financial depreciation rules may differ from operational lifecycle rules.

25.3 Internal Cost Allocation

Assets may generate internal project charges by:

- Hour
- Day
- Shift
- Kilometre

- Cycle
 - Fixed assignment fee
 - Usage unit
 - Percentage allocation
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26. Documents Integration

Every asset automatically receives a system folder in the Documents Module.

Recommended folder structure:

- 01 Asset Identity
- 02 Purchase and Procurement
- 03 Ownership and Registration
- 04 Technical Documentation
- 05 Manuals
- 06 Warranty
- 07 Insurance
- 08 Inspections
- 09 Maintenance
- 10 Repairs
- 11 Photos
- 12 Assignments and Transfers
- 13 Incidents and Claims
- 14 Financial Records
- 15 Disposal
- Transit

Documents may appear through references in:

- Asset folder
- Project folder
- Company folder
- Procurement folder
- Finance folder
- Maintenance folder
- Employee folder

The underlying document must not be duplicated.

27. Tasks Integration

The system must automatically create or link tasks for:

- Maintenance due
- Inspection due
- Insurance renewal

- Registration renewal
 - Warranty expiry
 - Return overdue
 - Missing documents
 - Defect correction
 - Transfer acceptance
 - Disposal approval
 - Meter reading
 - Asset count
 - Compliance issue
-

28. Calendar Integration

Calendar events may be generated for:

- Maintenance
 - Inspections
 - Reservations
 - Deliveries
 - Returns
 - Renewals
 - Certification expiry
 - Service appointments
 - Asset transfers
-

29. Employee Integration

Employee profiles must show:

- Assigned assets
- Asset custody history
- Current checkouts
- Overdue returns
- Driver or operator licences
- Asset-related certifications
- Asset incidents
- Damage responsibility
- Handover documents
- Assigned tools
- Assigned IT equipment

The system must prevent assignment where required qualifications are missing or expired.

30. Suppliers and Service Providers

Asset-related partners may include:

- Supplier
- Manufacturer
- Dealer
- Maintenance provider
- Repair workshop
- Inspection body
- Insurance provider
- Leasing company
- Rental company
- Transport provider
- Calibration provider

Provider performance metrics:

- Response time
 - Repair duration
 - Cost
 - Rework rate
 - Warranty recovery
 - Service quality
 - Availability
 - Compliance
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31. Disposal and Retirement

31.1 Disposal Methods

- Sale
- Auction
- Scrap
- Donation
- Return to supplier
- Lease return
- Write-off
- Transfer outside group
- Destruction
- Lost permanently

31.2 Disposal Workflow

1. Disposal proposal
2. Technical assessment
3. Financial assessment
4. Document verification

5. Approval
6. Asset lock
7. Sale or disposal process
8. Payment confirmation
9. Handover
10. Accounting closure
11. Document archive
12. Asset retirement

Disposed assets must remain searchable but read-only.

32. Asset Counting and Reconciliation

The system must support periodic asset audits.

32.1 Count Campaign

- Company
- Location
- Project
- Asset categories
- Responsible team
- Start date
- Deadline
- Scan method
- Count status

32.2 Count Results

- Found
 - Not found
 - Wrong location
 - Wrong custodian
 - Damaged
 - Unregistered asset
 - Duplicate record
 - Identifier unreadable
 - Requires investigation
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33. Mobile Architecture

The Assets Module must be fully usable on mobile browsers.

Critical mobile actions:

- Scan QR code

- View asset
- Check out
- Check in
- Transfer
- Record meter
- Report defect
- Add photo
- Complete inspection
- Start maintenance
- Finish maintenance
- Confirm receipt
- View documents
- Request asset
- Reserve asset

Offline capability should be supported for:

- Inspections
- Meter readings
- Photos
- Defect reports
- Checklists
- Signatures

Offline records must sync when connection is restored.

34. QR, Barcode, NFC, RFID, GPS, and IoT

34.1 MVP

- QR codes
- Barcodes
- Manual location
- Manual meter readings

34.2 Future Architecture

- NFC
- RFID
- GPS tracking
- Geofencing
- Engine telemetry
- Fuel sensors
- Temperature sensors
- Battery monitoring
- Predictive maintenance
- Automatic meter capture
- Usage analytics

External integrations must remain optional and should not be required for MVP operation.

35. Permissions Architecture

Permissions must be controlled by:

- Corporate group
- Legal company
- Department
- Project
- Location
- Asset category
- Asset record
- User role
- Workflow role

35.1 Recommended Roles

- Platform Administrator
- Group Asset Director
- Company Asset Manager
- Fleet Manager
- Equipment Manager
- Warehouse Manager
- Maintenance Manager
- Maintenance Technician
- Project Manager
- Site Manager
- Finance User
- Procurement User
- HSE User
- Employee/Custodian
- Auditor
- Read-only User

35.2 Permission Actions

- View
- Create
- Edit
- Delete draft
- Activate
- Assign
- Transfer
- Reserve
- Check out
- Check in
- Inspect
- Create maintenance order

- Approve maintenance
- Record costs
- View financial values
- Change ownership
- Dispose
- Archive
- Export
- Manage templates
- Manage classifications

Sensitive financial information must have separate permissions.

36. Approval Architecture

Configurable approval workflows must support:

- Asset purchase
- Asset registration
- Asset assignment
- Intercompany transfer
- High-value repair
- External maintenance
- Rental extension
- Disposal
- Sale
- Write-off
- Insurance claim
- Ownership change

Approval rules may depend on:

- Asset value
 - Company
 - Asset category
 - Project
 - Cost
 - Country
 - Risk level
 - User role
-

37. Notification Architecture

Notifications must support:

- In-app
- Email
- Mobile push when available

- Dashboard alert

Events include:

- Assignment request
- Assignment approval
- Reservation conflict
- Return due
- Return overdue
- Maintenance due
- Inspection due
- Compliance expiry
- Insurance expiry
- Warranty expiry
- Defect reported
- Critical failure
- Asset moved
- Asset missing
- Approval required
- Document missing
- Asset utilisation below threshold

Escalation rules must be configurable.

38. Dashboard Architecture

38.1 Executive Dashboard

- Total asset value
- Asset count
- Active assets
- Available assets
- Assets under repair
- Assets with expired compliance
- Maintenance overdue
- Asset utilisation
- Downtime
- Total maintenance cost
- Total operating cost
- Replacement exposure
- Lost and stolen assets

38.2 Operations Dashboard

- Available today
- Assigned today
- Reservations
- Transfers in progress
- Assets due back

- Defects
- Inspections due
- Work orders
- Location map

38.3 Maintenance Dashboard

- Work orders by status
- Preventive maintenance compliance
- Mean time between failure
- Mean time to repair
- Downtime
- Maintenance cost
- Repeated failures
- Parts consumption

38.4 Finance Dashboard

- Acquisition value
- Current book value
- Depreciation
- Total cost of ownership
- Cost by company
- Cost by project
- Cost by category
- Assets near replacement

39. Reporting Architecture

Required reports:

- Asset register
- Asset ownership report
- Asset assignment report
- Asset location report
- Asset movement history
- Asset utilisation report
- Idle asset report
- Asset cost report
- Maintenance history
- Maintenance cost report
- Downtime report
- Compliance expiry report
- Insurance expiry report
- Warranty expiry report
- Asset condition report
- Asset audit discrepancies
- Fuel consumption report
- Meter report

- Project asset allocation
- Employee custody report
- Disposal report
- Depreciation report
- Total cost of ownership
- Replacement forecast
- Lost and stolen assets
- Supplier service performance

Reports must support:

- Filters
- Saved views
- Scheduled delivery
- PDF export
- Spreadsheet export
- Role-based visibility

40. Search and Filtering

Global search must support:

- Asset code
- Asset name
- Serial number
- Registration number
- VIN
- Barcode
- QR code
- Manufacturer
- Model
- Employee
- Project
- Location
- Supplier
- Document
- Custom fields

Filters must include:

- Company
- Category
- Type
- Status
- Condition
- Availability
- Ownership
- Project
- Location

- Custodian
 - Maintenance status
 - Compliance status
 - Purchase date
 - Value range
 - Warranty status
-

41. Core Data Model

41.1 Main Entities

- Asset
- AssetCategory
- AssetType
- AssetModel
- AssetComponent
- AssetIdentifier
- AssetCustomField
- AssetCustomFieldValue
- AssetOwnership
- AssetAssignment
- AssetReservation
- AssetTransfer
- AssetLocation
- AssetLocationHistory
- AssetStatusHistory
- AssetConditionHistory
- AssetMeter
- AssetMeterReading
- AssetInspection
- AssetInspectionTemplate
- AssetInspectionResponse
- AssetDefect
- MaintenancePlan
- MaintenanceSchedule
- MaintenanceWorkOrder
- MaintenanceWorkLog
- MaintenancePartUsage
- AssetWarranty
- AssetInsurance
- AssetComplianceRecord
- AssetLicence
- AssetFuelTransaction
- AssetIncident
- AssetDamageRecord
- AssetFinancialRecord
- AssetDepreciationRecord
- AssetValuation
- AssetDisposal

- AssetAuditCampaign
- AssetAuditResult
- AssetDocumentReference
- AssetTaskReference
- AssetProjectReference
- AssetCostAllocation
- AssetNotificationRule

41.2 Critical Relationships

- Asset belongs to one responsible company
- Asset may have multiple ownership records over time
- Asset may have one active primary location
- Asset may have multiple historical locations
- Asset may have multiple assignments over time
- Asset may have multiple meters
- Asset may have multiple maintenance plans
- Asset may have multiple linked projects
- Asset may contain multiple component assets
- Asset may have one parent asset
- Asset may have many documents by reference
- Asset may have many financial transactions
- Asset may have many inspections and defects

42. Asset API Architecture

Recommended API groups:

- /assets
- /asset-categories
- /asset-types
- /asset-models
- /asset-components
- /asset-identifiers
- /asset-assignments
- /asset-reservations
- /asset-transfers
- /asset-locations
- /asset-meters
- /asset-inspections
- /asset-defects
- /maintenance-plans
- /maintenance-work-orders
- /asset-compliance
- /asset-warranties
- /asset-insurance
- /asset-costs
- /asset-depreciation

- `/asset-disposals`
- `/asset-audits`
- `/asset-reports`

All write APIs must support:

- Permission validation
 - Tenant validation
 - Company validation
 - Audit logging
 - Workflow validation
 - Optimistic concurrency
 - Idempotency where required
 - Attachment references
 - Historical record generation
-

43. Event Architecture

Recommended platform events:

- AssetCreated
- AssetActivated
- AssetUpdated
- AssetAssigned
- AssetReturned
- AssetReserved
- AssetTransferred
- AssetLocationChanged
- AssetStatusChanged
- AssetConditionChanged
- AssetInspectionCompleted
- AssetInspectionFailed
- AssetDefectReported
- AssetMaintenanceDue
- AssetMaintenanceStarted
- AssetMaintenanceCompleted
- AssetComplianceExpiring
- AssetComplianceExpired
- AssetWarrantyExpiring
- AssetMeterUpdated
- AssetDamaged
- AssetLost
- AssetRecovered
- AssetDisposed

These events may trigger tasks, notifications, financial allocations, approvals, or document creation.

44. Automation Rules

Examples:

- Create maintenance work order when service threshold is reached
 - Prevent assignment when inspection is expired
 - Prevent operation when critical defect is open
 - Notify manager before insurance expiry
 - Create return task before reservation end
 - Escalate overdue asset return
 - Update availability after assignment
 - Update location after accepted transfer
 - Create asset folder after activation
 - Create depreciation schedule after financial activation
 - Create defect after failed inspection
 - Lock asset after unsafe inspection
 - Recommend replacement when lifecycle threshold is reached
-

45. Audit and Governance

Every critical action must record:

- User
- Date and time
- Previous value
- New value
- Reason
- Source
- Device
- IP address where permitted
- Workflow approval
- Linked document

The platform must not allow deletion of active historical records.

Correction must occur through:

- Reversal
 - Amendment
 - Replacement record
 - Approved correction workflow
-

46. Security and Data Segregation

The module must follow the Construction OS hybrid tenancy architecture.

Required controls:

- Tenant isolation
 - Corporate group isolation
 - Legal entity boundaries
 - Project-level access
 - Record-level permissions
 - Financial field restrictions
 - Personal data minimisation
 - Attachment access control
 - Audit logging
 - Export control
 - API access control
-

47. User Interface Architecture

47.1 Main Navigation

Assets

Sub-navigation:

- Overview
- All Assets
- My Assets
- Available
- Assigned
- Reservations
- Transfers
- Maintenance
- Inspections
- Compliance
- Locations
- Asset Audit
- Reports
- Settings

47.2 Assets List

Recommended columns:

- Asset
- Asset code
- Category
- Company
- Project
- Location
- Custodian

- Status
- Condition
- Availability
- Next maintenance
- Compliance
- Value
- Actions

47.3 Asset Detail Tabs

- Overview
- Technical
- Assignments
- Reservations
- Location
- Maintenance
- Inspections
- Compliance
- Meters
- Costs
- Documents
- Incidents
- History

48. Settings Architecture

Administrators must manage:

- Asset classifications
 - Asset code formats
 - Custom fields
 - Statuses
 - Condition scales
 - Location types
 - Assignment rules
 - Reservation rules
 - Maintenance templates
 - Inspection templates
 - Meter types
 - Compliance types
 - Depreciation methods
 - Cost rates
 - Approval workflows
 - Notification rules
 - QR label templates
 - Disposal reasons
 - Document requirements
 - Import mappings
-

49. Import and Migration

The system must support import of:

- Existing asset register
- Serial numbers
- Historical assignments
- Current locations
- Purchase values
- Depreciation data
- Maintenance history
- Meter readings
- Insurance
- Compliance documents
- Employee assignments

Import process:

1. Upload file
2. Map fields
3. Validate data
4. Detect duplicates
5. Preview
6. Correct errors
7. Approve import
8. Create records
9. Generate import report

50. Non-Functional Requirements

- Multi-company support
 - Multi-currency support
 - Multi-language support
 - Time-zone support
 - Mobile responsive
 - Fast global search
 - Bulk actions
 - Full audit trail
 - High-volume asset support
 - Configurable workflows
 - API-first architecture
 - Secure file references
 - Offline-capable inspections
 - Scalable event processing
 - Role-based data access
-

51. MVP Scope

The first implementation should include:

- Asset register
 - Asset categories and types
 - Asset identity
 - Ownership
 - Locations
 - Assignments
 - Check-in/check-out
 - Transfers
 - QR codes
 - Asset documents
 - Asset status and condition
 - Maintenance plans
 - Maintenance work orders
 - Inspections
 - Defects
 - Compliance expiry tracking
 - Basic financial values
 - Project links
 - Employee links
 - Dashboard
 - Reports
 - Audit trail
 - Notifications
-

52. Phase Two

- Reservations
 - Fuel management
 - Depreciation engine
 - Internal asset charging
 - Advanced asset audits
 - Warranty claims
 - Insurance claims
 - Offline workflows
 - Advanced supplier performance
 - Asset replacement forecasting
-

53. Future Roadmap

- GPS
- IoT telemetry
- RFID

- NFC
- Predictive maintenance
- Automated fuel data
- Digital twins
- AI defect analysis
- AI maintenance recommendations
- Image-based asset recognition
- Telematics integrations
- External accounting integrations
- External fleet platforms
- Manufacturer integrations

These integrations must remain outside the initial self-contained MVP.

54. Final Architectural Position

The Assets Module must not operate as a simple equipment list.

It must function as the complete enterprise asset lifecycle system for Construction OS.

It connects:

- Procurement
- Inventory
- Projects
- Employees
- Documents
- Tasks
- Calendar
- Finance
- Maintenance
- HSE
- Quality
- Site Operations
- Suppliers
- Approvals
- Reporting

The permanent core principle is:

One asset, one permanent record, one complete lifecycle, and one auditable history across the entire organisation.